● v0.9.15 · updated 5 Nov 2026, 19:52 UTC

FM-STEP

A synth groovebox that runs on the Akai MPC Sample.

Eight tracks of synthesis, a 64-step sequencer and live performance effects, launched from the SD card next to the stock firmware. Nothing is flashed; the normal MPC is always one exit away.



Interface illustration · EQ page over the step view v0.9.1

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01 · The idea

A sampler and a groovebox in one box

Most mods for the MPC Sample extend it as a sampler. FM-STEP turns the same hardware into a synth groovebox: sound engines you play and sequence step by step, with an interface built for jamming live. The plan is a hybrid instrument, with sample tracks joining the synth tracks.

The MPC Sample as it ships

- Sampler engine with real-time sequencer, swing and automation
- No step sequencing
- No synth engine
- 16 pads, 3 knobs, encoder, 2.4" screen

Source: Synth Anatomy, Oct 2026

With FM-STEP running

- 8 synth tracks on 23 sound models, each with its own sound chain
- 16 patterns of up to 64 steps, with locks, swing, timing nudges and chains
- 8 fader performance effects and pad toggles for live jamming
- Live graphics for every parameter on the 2.4" screen
- Back to stock with SHIFT + STOP, STOP

02 · Features

What's in v0.9

Everything below runs on the MPC Sample hardware today, on firmware 1.3.0.12.

Sound

- 23 models from Mutable Instruments' Plaits (MIT): analog, FM, DX7 banks, chords, modal, drums
- Sub oscillator, drive ladder filter, tempo-synced wobble
- 3-band EQ per track, from +18 dB to a DJ-style kill
- Dub tape delay with throws, space echo, reverb
- Warmth and tape: drift, chorus, glue, saturation, hiss, crackle

Sequencer

- 8 tracks, each drum on its own track
- 16 patterns × up to 64 steps, chains of up to 8
- Pitch, velocity, length and ±45% timing per step
- Up to 4 parameter locks per step; live recording of notes and knob moves
- Swing, quantise, mutes, live length pads that stay on the beat

Performance

- Fader effects: low pass, high pass, slicer, stutter, space echo, reverb wash, pan, volume
- Pick any effect from any screen; switch each on or off from the pads
- One-press reset of any knob to neutral
- Two-step STOP: let the echoes ring, or silence everything

Interface

- Its own drawing engine: anti-aliased curves, arc knobs, dot-matrix numbers
- Live EQ, filter, envelope, delay, reverb and tape graphs
- Dot-grid steps: size and brightness show velocity, capsules show long notes
- Pads and views follow the playhead; instructions are built into the screen

Safety

- Launched from the SD card; nothing flashed or overwritten
- Firmware check and file integrity check before launch
- Watchdog, crash restart, and a breaker that falls back to stock
- Projects autosaved to the card every 30 s

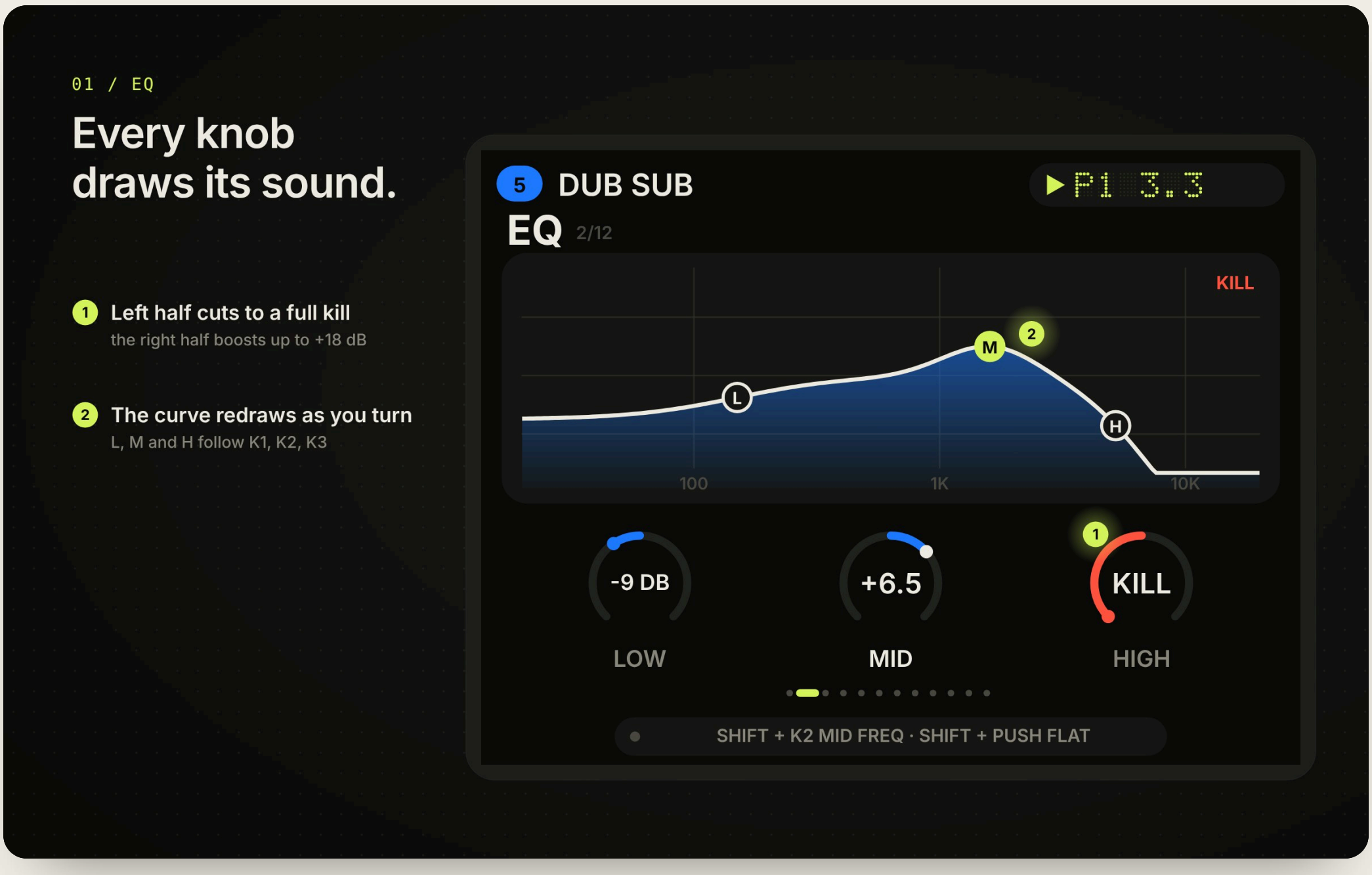
Extras

- Audio over USB-C to a computer or phone (works; level still being tuned)
- 16 project slots and 4 sound banks on the card
- Offline renderer for auditioning the engine on a computer

03 · Screens

Designed for a 2.4-inch screen

Illustrations of the interface, drawn from the app's own layouts on the MPC's 320 × 240 screen. Every page pairs a live graphic with three arc knobs for K1–K3. Videos of it running on the MPC are coming.

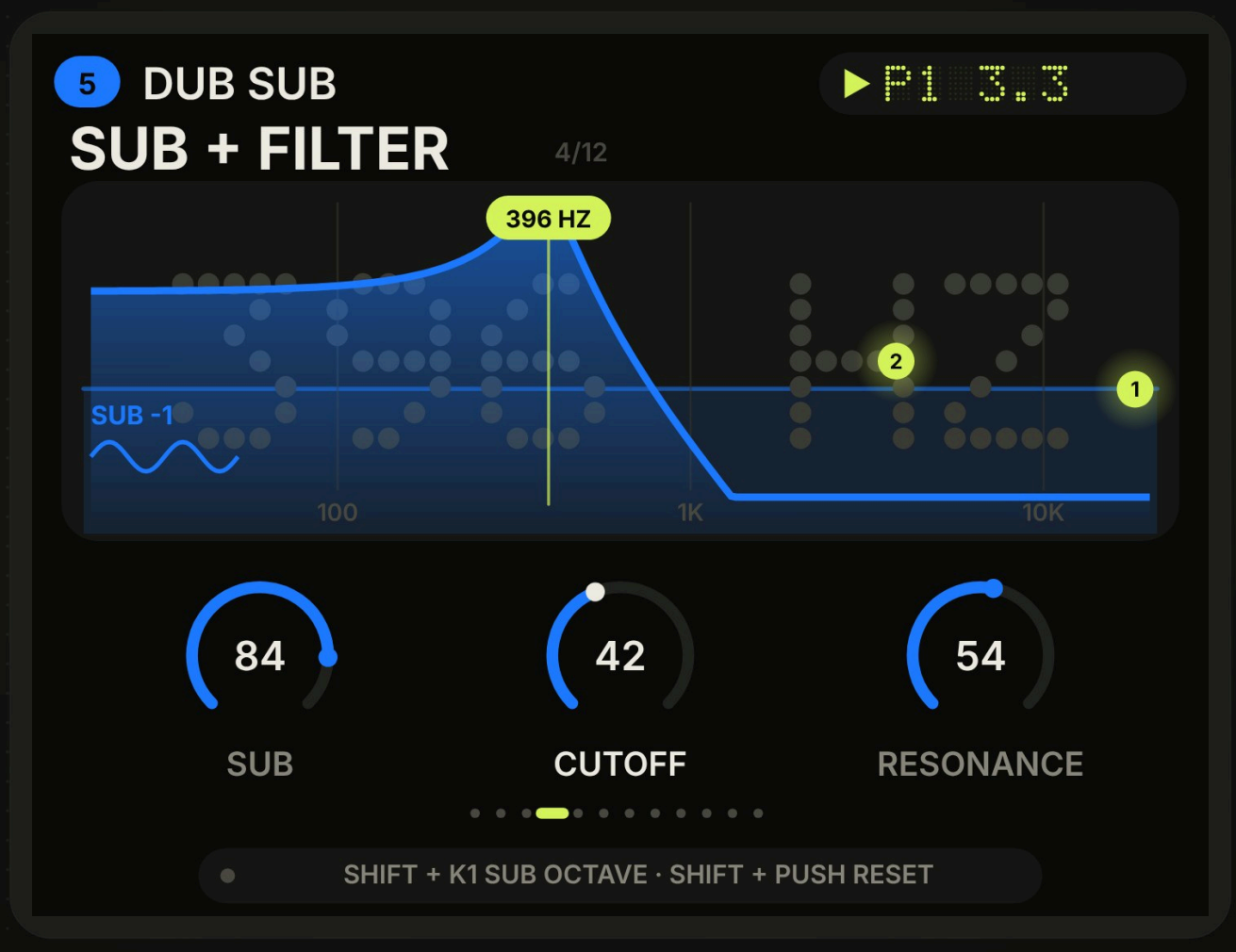


Every knob draws its sound. The EQ curve redraws as you turn; the left half of each knob cuts all the way to a kill.

02 / LIVE LAYER

You see the change happen.

1. **The panel fills to the value**
   in the track colour, then fades
2. **The number, huge and faint**
   behind the graph, in dot matrix

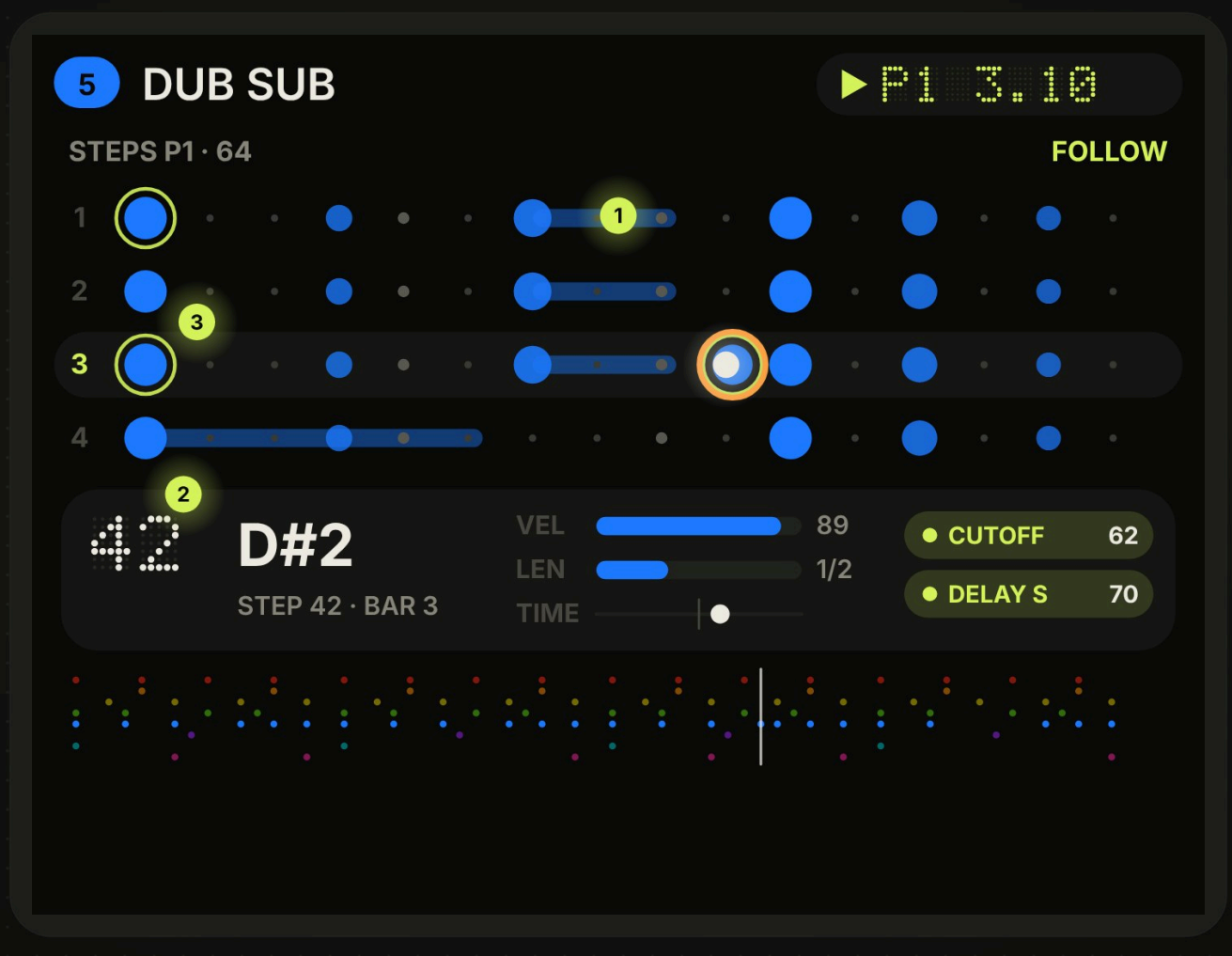


You see the change happen. While a knob turns, the panel fills to its value and the number appears huge and faint behind the graph, then fades.

03 / 64 STEPS

Four bars, at a glance.

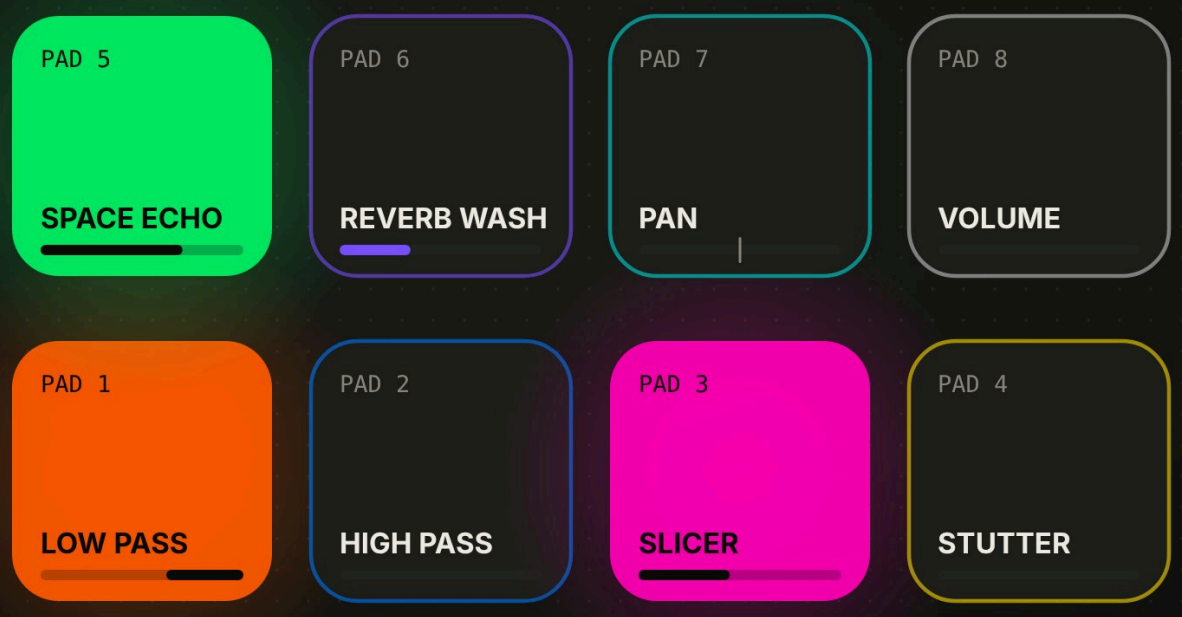
1. **Long notes are capsules**
   dot size is velocity
2. **Hold a step to edit it**
   note, velocity, length, timing, locks
3. **Lime rings: parameter locks**



Four bars at a glance. 64 steps as dots: size is velocity, capsules are long notes, lime rings are parameter locks. Hold a step to edit it.

04 / PERFORMANCE

Eight effects under your fingers.



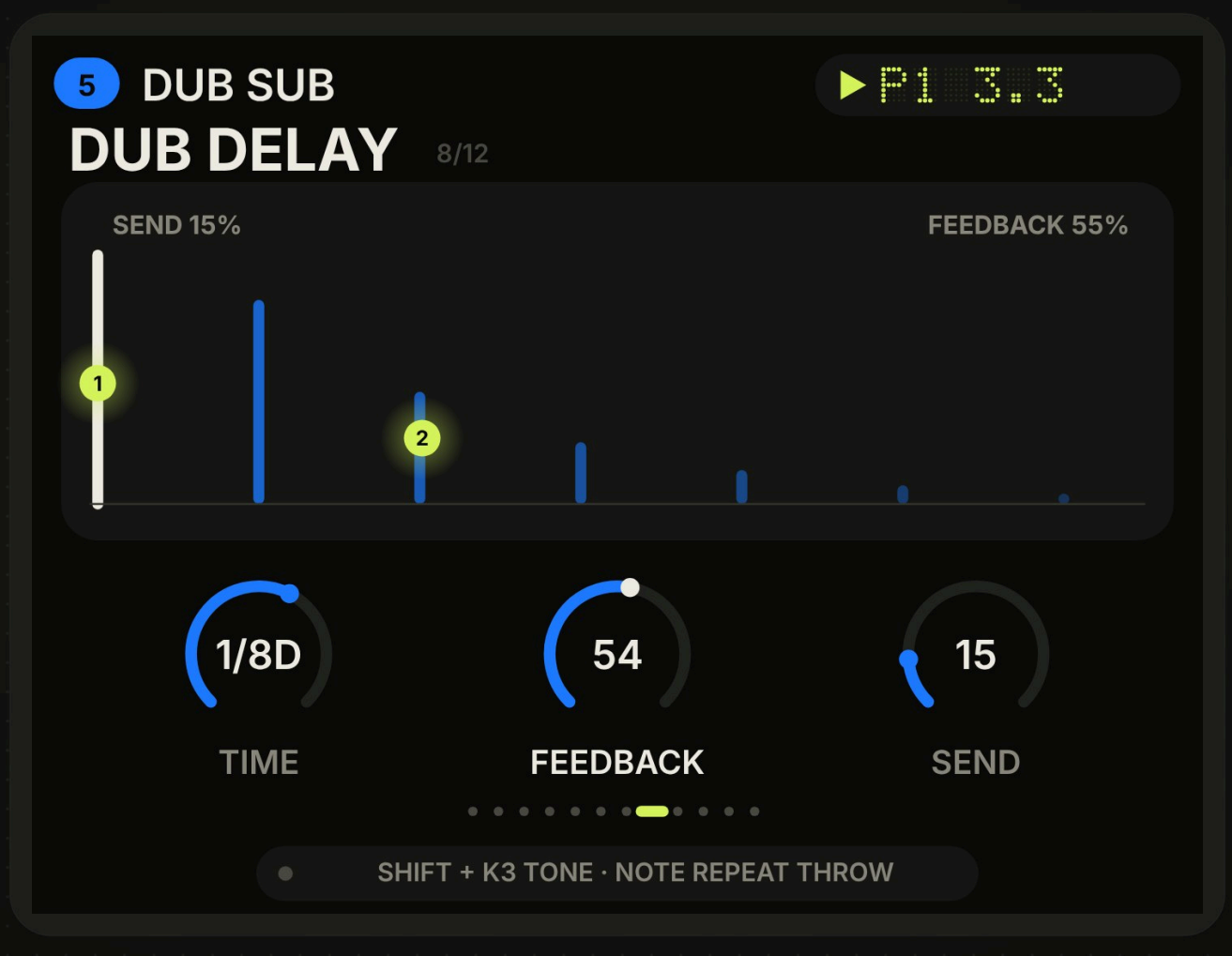
Tap a pad to switch an effect on or off, from any page.
The fader rides whichever one you picked; the bars show how much.

Eight effects under your fingers. Low pass, high pass, slicer, stutter, space echo, reverb wash, pan and volume, switched from the pads on any page.

05 / DUB DELAY

Echoes you can read.

1. The dry hit
2. Each echo, shrinking by feedback
   time synced to the tempo

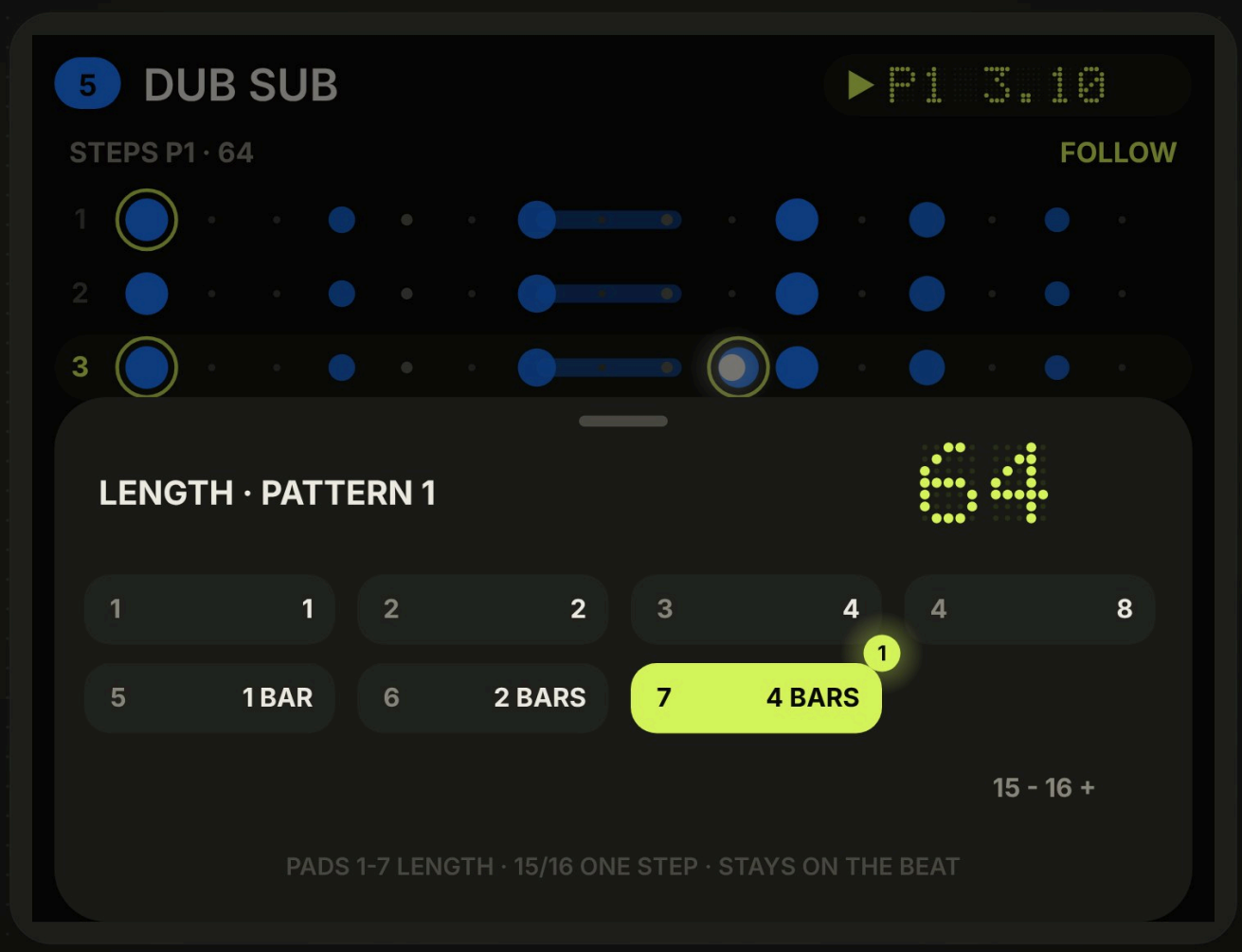


Echoes you can read. Each repeat is a bar, shrinking by feedback, in time with the tempo.

06 / LIVE LENGTH

Change the loop mid-beat.

1. **Hold LOOP, tap a pad**
   1 step up to 4 bars



Change the loop mid-beat. Hold LOOP and tap a pad: 1 step up to 4 bars, without losing the beat.

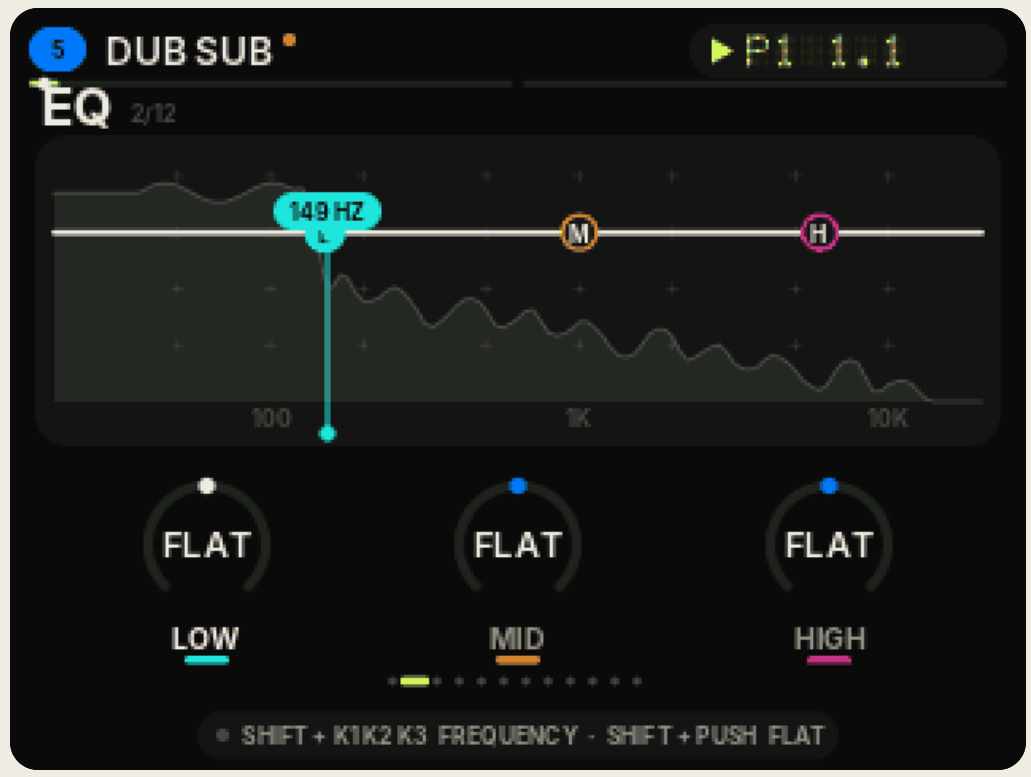
03b · In motion

How it moves

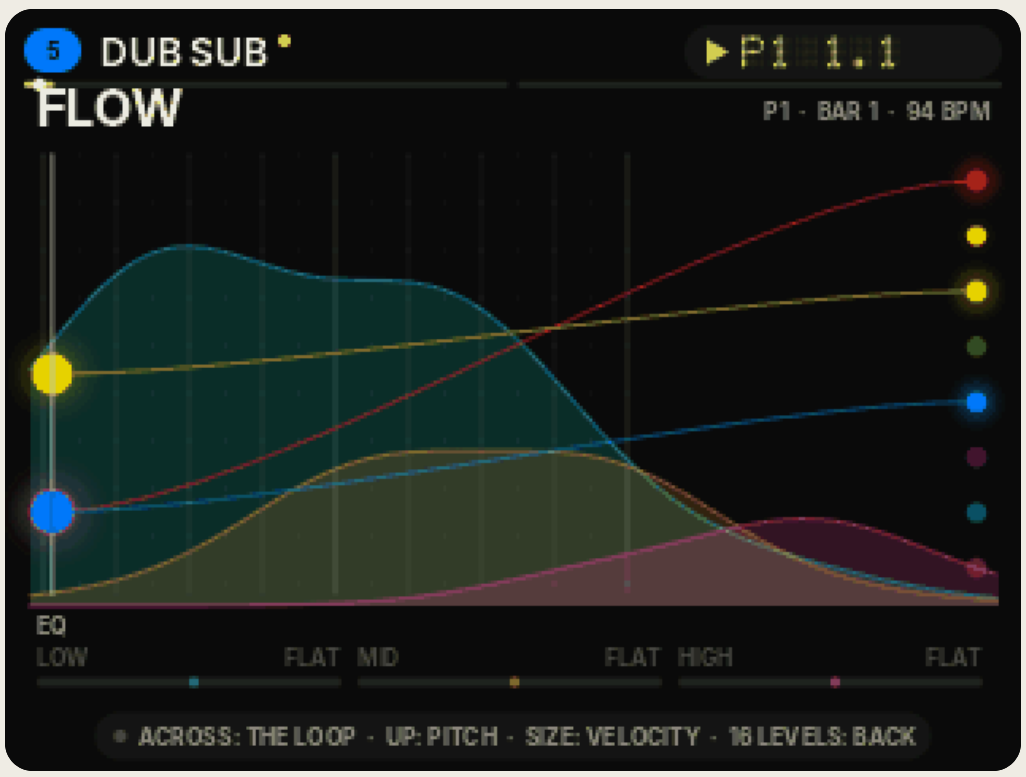
Short loops drawn by the app's own code at the screen's 320 × 240, shown at 2×. The sound data is synthetic so the loops repeat cleanly; on the MPC everything follows the music.



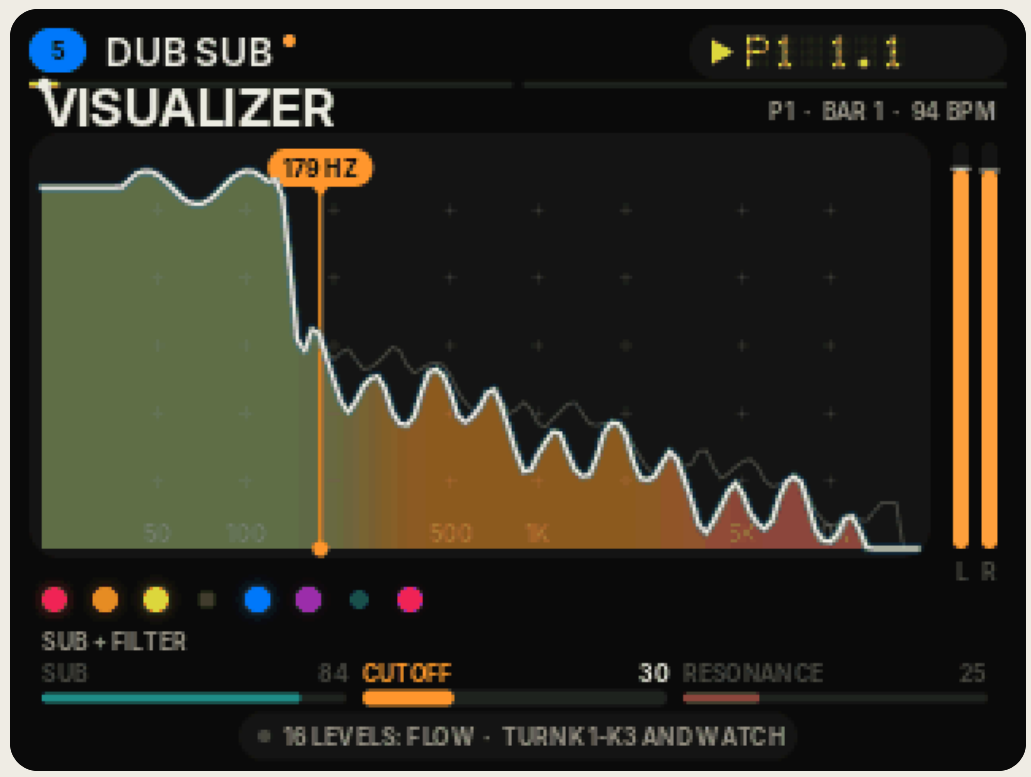
The knob flag. Turn a knob and a flag in its colour rides with the value, on every page and visualizer.



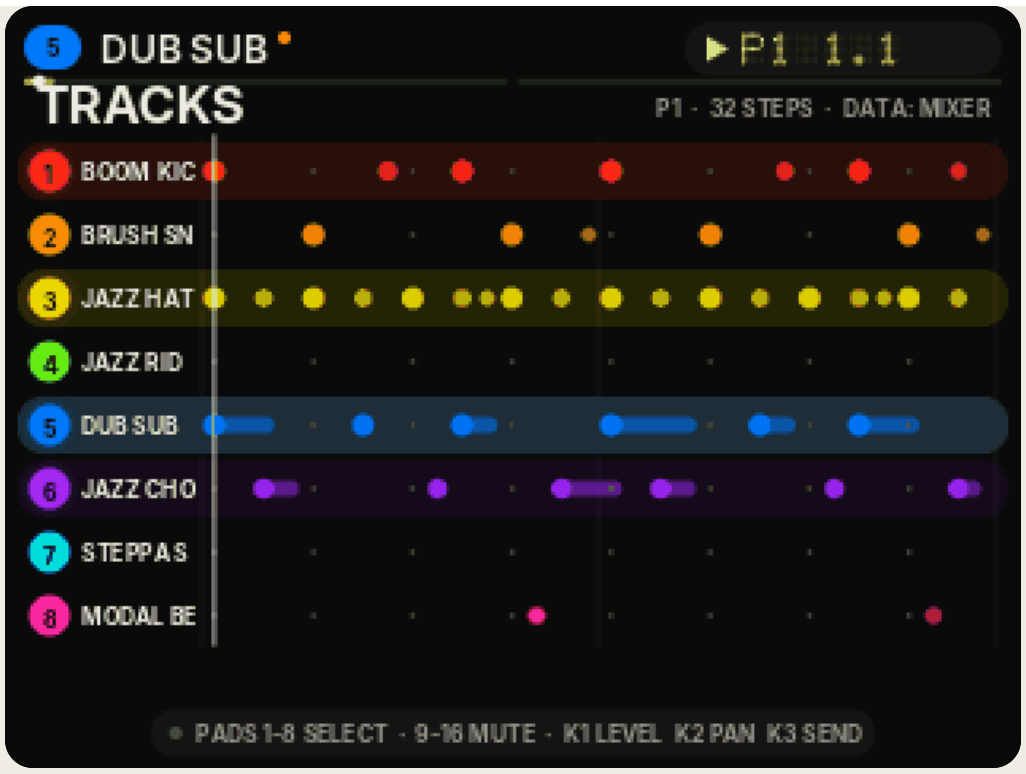
EQ over the live sound. The low band cuts to a kill and back; the track's spectrum behind it follows.



FLOW. Every note drops a bubble on the beat grid and flows to its track; the colour hills swell with the music.



SPECTRUM. The mix by frequency; turning CUTOFF washes it orange and flags the frequency.



All tracks. Every track's notes with the playhead and the loop line on top.

04 · Logs & reports

Tested on the hardware

Every version is written to the card, checked against its file hashes, and played on the MPC. The app and its supervisor log timing and recovery events, and those logs drive the fixes.

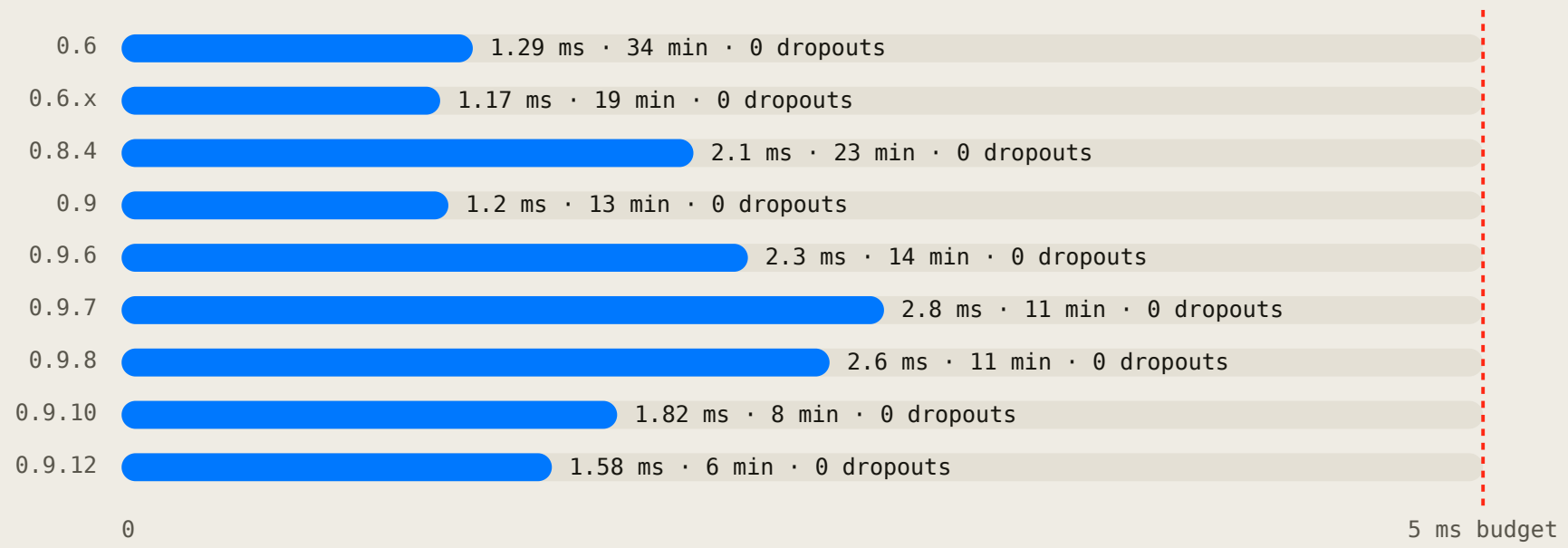
0
audio dropouts in the logged sessions (23 and 43 min)

2.1 ms
worst audio block, of a 5 ms budget

25/25
automated recovery tests

12.1 dB
mix dynamics after the gain fix (was 7.6 dB)

Worst audio block per logged session — how long the sound engine took for its slowest 5 ms block of audio on the MPC. Under the line means no dropouts.



TEST	VERSION	RESULT	DETAIL
Normal exit, app crash, app freeze, supervisor crash, SD card pulled, power cut during start, circuit breaker, cold boot without card	0.1–0.3	Passed	The stock MPC software returned every time
Long playback	0.8.4	Passed	~23 min with no dropouts; worst block 2.1 ms of 5 ms
Exit to stock during playback	0.8.5	Passed	Twice; the stock software restored cleanly
Forced power-off mid-jam, then restart	0.8.5	Passed	Normal boot; project loaded intact
Exit in the first 10 seconds counted as a failed launch	0.8.6	Fixed	Found in the logs; a deliberate exit no longer counts against the breaker
Mix compressed and distorted with 8 tracks	0.7.1	Fixed	Re-staged levels; dynamics from 7.6 to 12.1 dB with the tape character kept
Grainy reverb tails	0.7.2	Fixed	12-bit reverb memory replaced with full resolution; tails now fade smoothly
Pops when moving effects	0.7.3	Fixed	Clicks on a held tone: 9 before, 0 after
Audio over USB-C	0.8.4	Works	44.1 kHz with clock tracking, no skips
USB-C recording level too hot	0.8.5	Open	To be solved with the mixer and output engine

05 · Build log

The detailed build log

Every version that went onto the MPC's card, in order, with the date and time it was built. Open an entry for what changed, what we measured, and how it looks. Times are UTC: exact from 0.9.1 on (when the build was written to the card); entries marked ≈ come from when that stage was documented.

3 Nov 2026
21:32 UTC

`0.1–0.3` **Safe launch and recovery** — Played on the MPC

- Launched from the SD card next to the stock firmware: nothing flashed, firmware 1.3.0.12 checked, every file hash-checked before launch.
- Supervisor with heartbeat watchdog, crash restart and a circuit breaker that hands back to the stock MPC software.
- Recovery tests on the MPC: normal exit, app crash, app freeze, supervisor crash, SD card pulled, power cut during start, breaker, cold boot without a card. All passed.
- Lab app mapped the controls: 16 pads with velocity and pressure, K1–K3, fader, DATA encoder and 22 buttons.

4 Nov 2026
≈ 13:50 UTC

`0.4` **First sound** — Played on the MPC

- A four-operator FM voice played from the pads with velocity, 8 algorithms, ADSR per operator, presets.
- Found and fixed a launch failure (UI blocked on a control read) the same day; regression test added.

4 Nov 2026
≈ 18:00 UTC

`0.5–0.6` **Plaits engines, dub layer, warmth and tape** — Played on the MPC

- 23 sound models from Mutable Instruments' Plaits (MIT) with a sub oscillator, drive ladder filter, wobble, dub tape delay and reverb.
- Character chain: analog drift, softness, chorus, tape glue, saturation, hiss and crackle.
- Fader performance effects and exact tap tempo.
- First long session: 34 min, 0 dropouts, worst audio block 1.29 ms of 5 ms.

4 Nov 2026
≈ 23:10 UTC

`0.7` **The sequencer** — Played on the MPC

- 8 tracks, 16 patterns, chains, real-time recording of notes and knob moves, mutes.
- Steps with pitch, velocity, length, timing nudges and parameter locks.

5 Nov 2026
overnight UTC

`0.7.1–0.8.6` **Sound fixes, 64 steps, EQ, USB-C audio, reliability** — Played on the MPC

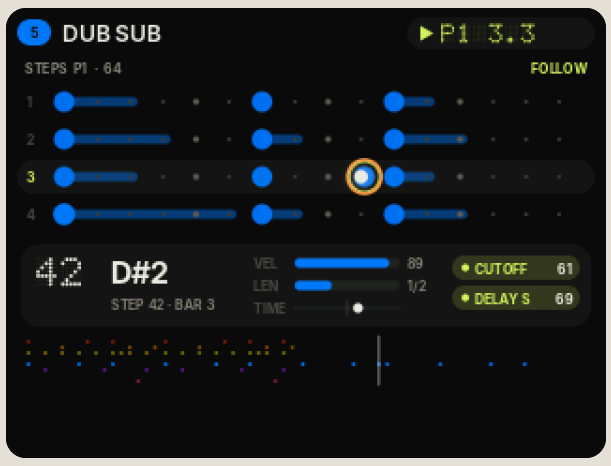
- Gain staging rebuilt: mix dynamics from 7.6 to 12.1 dB. Full-resolution reverb. Pops fixed (held-tone clicks 9 → 0).
- 64-step patterns, pads follow the playhead, live length pads, per-track 3-band EQ with a DJ kill, one-press resets, effect toggles, two-step STOP.
- Audio over USB-C with clock tracking (works; level still too hot).
- Reliability on the MPC: ~23 min playback with 0 dropouts, exit to stock during playback, forced power-off with the project intact.

5 Nov 2026
≈ 07:00 UTC

0.9 New interface

Played on the MPC

- Own drawing engine: anti-aliased curves and arcs, Inter type, dot-matrix numbers, off-screen drawing.
- Every parameter page gets a live graphic and three arc knobs; the step view becomes a dot grid with long-note capsules, lock rings and a held-step card.
- 13 min on the MPC with 0 dropouts.



Step view: four bars of dots, a held step's card with note, velocity, length, timing and locks. (preview render)

5 Nov 2026
07:40 UTC

0.9.1 Faster screen, bigger page names

Played on the MPC

- Logs showed each full-screen update took 54 ms: the screen now receives only the part that changed, and unchanged frames cost nothing (average 2–18 ms).
- Page names large; page switches instant with only the title sliding in (no more ghosting).
- First live layer while turning a knob.

5 Nov 2026
07:59 UTC

0.9.2 One colour per knob

Played on the MPC

- K1 cyan, K2 orange, K3 magenta, with a matching tag under each knob; model names flash large in yellow dot matrix.

5 Nov 2026
08:11 UTC

0.9.3 Knob heat map

Played on the MPC

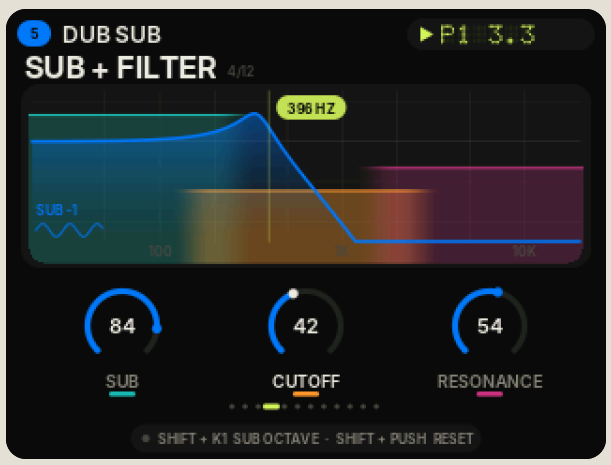
- All three knob settings always visible as overlapping bands of light behind the graph.

5 Nov 2026
08:24 UTC

0.9.4 Watercolour washes

Played on the MPC

- The heat map banded into a mosaic on the panel; redrawn as flat, soft-edged washes that blend where they overlap.



Sub + filter with the three knob washes while CUTOFF turns. (preview render)

5 Nov 2026
08:36 UTC

0.9.5

Every view redesigned

Played on the MPC

- Mixer strips, a pattern grid with mini maps and chain order, effect pads, projects, sound banks and an exit card, all in the new style.



Patterns: tiles laid out like the pads, each with a map of its notes; the playing one glows. (preview render)

5 Nov 2026
08:48 UTC

0.9.6

Live spectrum

Played on the MPC

- The sound engine measures the music 25 times a second in 48 bands.
- EQ bands drawn as layered colour shapes over the spectrum; a full visualizer on 16 LEVELS.
- On the MPC: 14 min, 0 dropouts, worst block 2.3 ms.



EQ: each band's shape in its knob colour over the live spectrum, combined curve in white. (preview render)

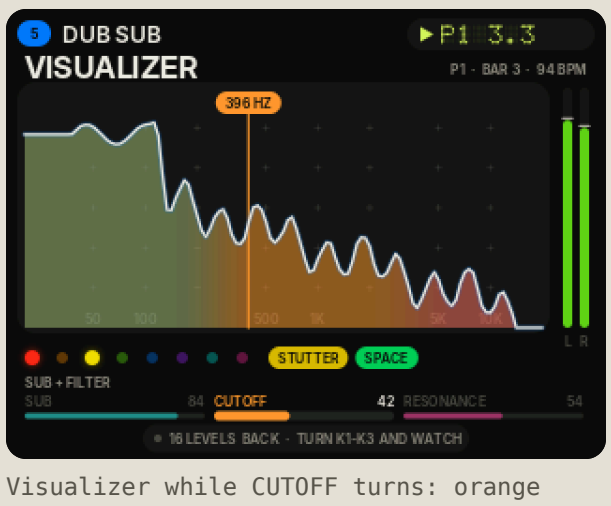
5 Nov 2026
09:22 UTC

0.9.7

Track spectrum, knob flags, dot fields

Played on the MPC

- The EQ page shows the selected track's own sound; the low kill now removes the sub too (−12 → −46 dB below 90 Hz).
- On the visualizer, turning a knob washes the spectrum in its colour and flags its frequency.
- A slow dot field glows with the sound behind every page.



Visualizer while CUTOFF turns: orange wash and a 396 Hz flag. (preview render)

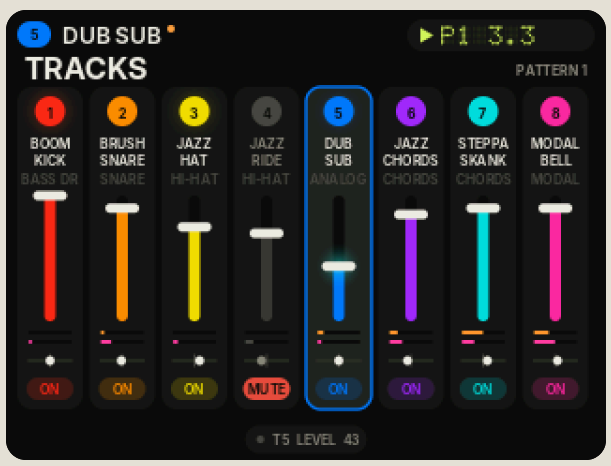
5 Nov 2026
09:47 UTC

0.9.8

Mixer knobs, a different motion per page

Played on the MPC

- K1 level, K2 pan, K3 sends on the mixer.
- Dot motion per page: ripples, rain, waves, twinkle, mirror.
- On the MPC: 11 min, 0 dropouts, worst block 2.6 ms; button log mapped which buttons send.



Mixer while K1 sets track 5's level. (preview render)

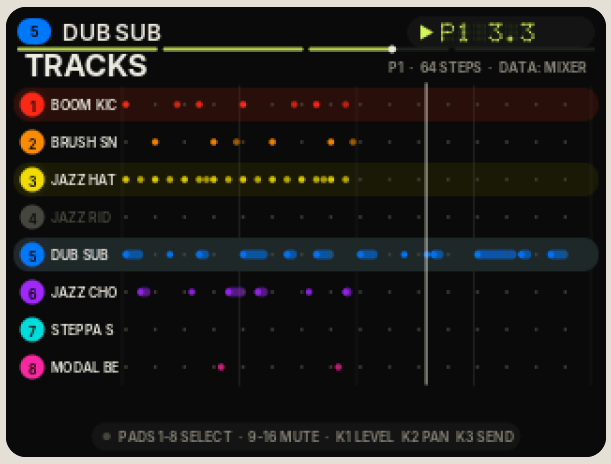
5 Nov 2026
13:39 UTC

0.9.9

Loop line, notes view, erase all

Played on the MPC

- A loop line under the header on every screen, one segment per bar.
- The all-tracks notes view is back beside the mixer.
- SHIFT + ERASE twice clears a track or a pattern; a third press undoes.



Tracks: every track's notes, the playhead, rows glowing on hits, and the loop line on top. (preview render)

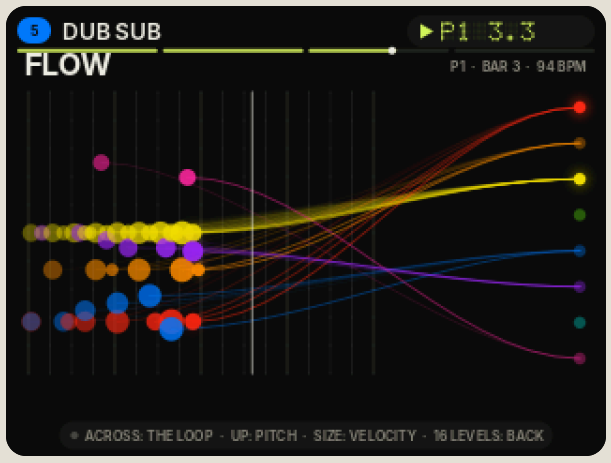
5 Nov 2026
13:48 UTC

0.9.10

FLOW visualizer

Played on the MPC

- 16 LEVELS now cycles pages → spectrum → flow.
- Every note drops a bubble on a beat grid (loop position, pitch, velocity, track colour) and flows to its track's lane.
- On the MPC: 8 min, 0 dropouts, worst block 1.8 ms. Button log: every button sends except the three screen buttons, which were not pressed in that session; PAD FX is the one free button.



FLOW: two bars of the starter groove. (preview render)

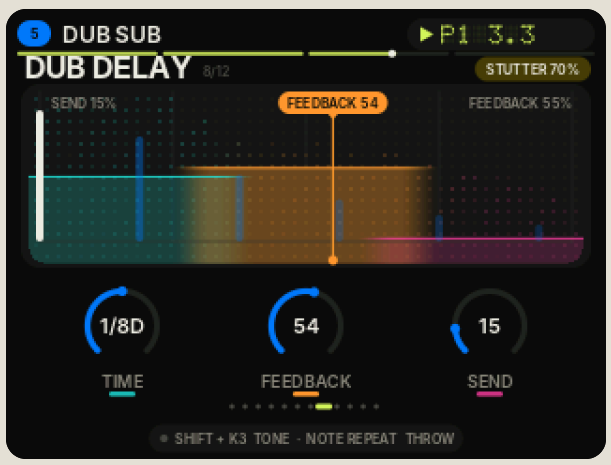
5 Nov 2026
14:07 UTC

0.9.11

Flags everywhere, watercolour FLOW

Played on the MPC

- The flag you liked on the visualizer now rides with every knob: on every parameter page, on SPECTRUM and on FLOW. It sits at the frequency for frequency settings and at the value for everything else, in the knob's colour, with the value on its chip.
- FLOW gets a watercolour layer: three soft hills of colour (lows cyan, mids orange, highs magenta) swell with the music and drift, blending where they overlap; the knob being turned deepens its hill.



Dub delay: the FEEDBACK flag rides with K2. (preview render)

5 Nov 2026
14:32 UTC

0.9.12

Screen buttons and a calmer LOOP

Played on the MPC

- The three screen buttons are confirmed working (notes 12, 13, 14). Left pages down, right pages up, and the middle button flips to the other face of the view: a page and its visualizer, the tracks' notes and the mixer, a followed or held bar in the step view.
- LOOP no longer does one thing on a tap and another on a second tap. Hold LOOP for the loop-length sheet and pads (it appears after a short hold, so nothing flashes); SHIFT + LOOP opens the patterns view; ERASE + LOOP clears the chain.
- Short check of 0.9.11 on the MPC: 0 dropouts, worst block 0.9 ms.
- On the MPC: 6 min, 0 dropouts, worst block 1.6 ms. The user approved the screen buttons and the middle-button faces.

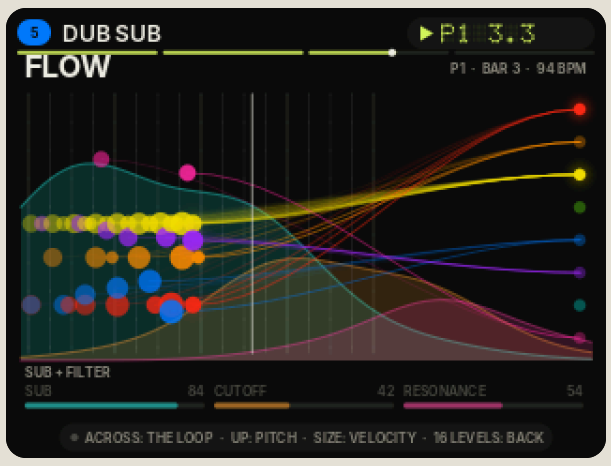
5 Nov 2026
16:46 UTC

0.9.13

EQ frequencies on all three bands, FLOW says what the knobs do

On the card, test pending

- EQ: SHIFT + K1 moves the low shelf (40–400 Hz) and SHIFT + K3 the high shelf (2–16 kHz), like SHIFT + K2 already moved the mid (200 Hz–5 kHz). Defaults stay at 150 Hz and 6 kHz, so existing sounds are unchanged. Offline check: a low kill at 150 Hz removes 15.6 dB in the 80–200 Hz band; moved down to 40 Hz it removes only 2.8 dB.
- FLOW now shows the page the knobs are on and K1–K3 as bars, like the spectrum view, so you always know what you are changing.
- PAD FX shows a note: GRAB and slicing arrive with FLUX part 2.



FLOW with the page and knobs along the bottom. (preview render)

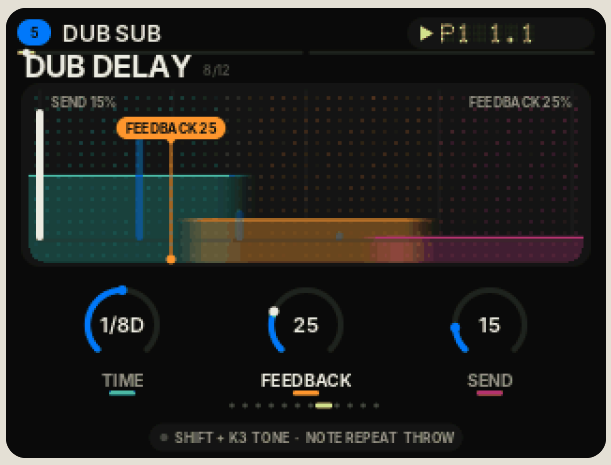
5 Nov 2026
19:52 UTC

0.9.15

First public beta

On the card, test pending

- Heavier small type (Inter Bold for small text and labels, slightly brighter grey) for readability on the 2.4-inch screen; flags on the parameter pages sit below the graph labels.
- Public repository prepared: MIT licence, install, controls, build and compatibility docs, a changelog, third-party notices, and the interface in motion as GIFs drawn by the app's own code.
- Reproducible builds: `make release` builds everything from source; the sound engine fetches Plaits, Clouds and stmlib at pinned commits. A from-scratch build rendered audio bit-identical to the binary tested on the MPC.
- Release zip ships OFF: players create FMSTEP/ENABLE to turn it on.



The knob flag, as a GIF for the repo. (drawn by the app)

Next

1.0: startup animation and release docs, then FLUX part 2: always-rolling tape, grab, and the motion slicer.

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06 · How it's made

Made by an artist, for playing out

FM-STEP is built by Darren Alexander Cole, a moving-image and sound artist studying computational media in the Interactive Media & Game Development (IMGD) department at Worcester Polytechnic Institute (WPI). Cole has more than 20 years of experience as an artist working between sound and moving image, from 16mm film projects to augmented and virtual reality experiences.

Made with the help of Claude, Anthropic's AI, this custom mod for the MPC Sample grows out of a lifelong love of exploring the roots of hip hop: jazz groove sessions and deep-listening modular sets that leave you wondering whether you just left Jamaica or are on a flight to Europe. Every change has been tested on the MPC, and a slow beta-testing phase will follow. FM-STEP is where live performance meets the kinetic energy of the sample-based producer's loop.

Engine

- C++ sound engine and sequencer clock in a real-time audio thread, 5 ms blocks

Interface

- Go app with its own drawing engine, Inter typeface and dot-matrix text

Safety

- Supervisor with watchdog and breaker; stock always one exit away

Thanks

Inspiration

GwalaKeyz

Mike Feedback

Jsen Surreal

Aleksey Malykh (am-mod)

Killa Vic

For listening

Kate Mytty

Jerrold Goodman (Jayroll)

Husani Lewis

Mike Wilson

5D5K

OMNI.SONO

Doctor Igor

Ben Olsen

Open source

Plaits and Clouds by Émilie Gillet / Mutable Instruments (MIT)

Inter by Rasmus Andersson (SIL OFL 1.1)

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